

2024044

B.E. II EXAMINATION DEC.' 2023
Civil Engg.
3AVRC1: Applied Mathematics-III

Time: 3 Hrs.]

[Max. Marks: 60]

Note: Attempt any TWO parts from each Question. Each carries equal marks.

Q.1. (a) Evaluate:

(i) $L\{t \cos at\}$

(ii) $L\left\{\int_0^t \frac{e^{-t} \sin t}{t} dt\right\}$

(06)

(b) Using convolution theorem, evaluate $L^{-1}\left\{\frac{1}{(s^2+a^2)^2}\right\}$.

(06)

(c) Solve by method of Laplace transform the equation $y''' + 2y'' - y' - 2y = 0$, given that $y(0) = 0$, $y'(0) = 0$, $y''(0) = 6$. (06)

Q.2. (a) 100 students of management institute obtained the following grade in statistics paper: (06)

Grade	A	B	C	D	E	Total
Frequency	15	17	30	22	16	100

Using χ^2 test, examine the hypothesis that the distribution of grades is uniform.
 Given that $\chi_4^2(0.05) = 9.49$.

(b) A certain stimulus administered to each of the 12 patients resulted the following increments of blood pressure: 6, 2, 8, -1, 8, 0, -2, 1, 5, 0, 5, 7. Can it be concluded that the stimulus has significant effect in increasing the blood pressure.
 (Take $t_{0.05} = 2.2$) (06)

(c) Explain Sampling Theory with Examples. (06)

Q.3. (a) Find the root of the equation $x \log_{10} x - 1.2 = 0$ by Regula falsi method correct upto three decimals. (06)

(b) Solve the following equations by Jacobi iterative method. (06)

$$20x + y - 2z = 17; \quad 3x + 20y - z = -18; \quad 2x - 3y + 20z = 25.$$

(c) Solve the differential equation $\frac{dy}{dx} = x + y^2$ by Runge's Kutta method with $y(0) = 1$ and find y at $x = 0.2$ in steps of 0.1. (06)Q.4. (a) Find the polynomial $f(x)$ by using Lagrange's formula and hence find $f(3)$: (06)

x	0	1	2	5
$f(x)$	2	3	12	147

(b) Find the value of $f(22)$ and $f(42)$ from the following available data: (06)

x	20	25	30	35	40	45
$f(x)$	354	332	291	260	231	204

(c) Evaluate $\int_0^1 \log \cos x \, dx$ using (a) Trapezoidal rule and (b) Simpson's one third rule. (06)

PTO

Q.5. (a) Find the Fourier series of the function $f(x) = \frac{x^2}{\lambda}$, for $-\pi$ to π . (06)

(b) Find the Fourier Transform of (06)

$$f(x) = 1 - x^2, |x| \leq 1$$

$$= 0, |x| \geq 1$$

Hence evaluate $\int_0^\infty \frac{x \cos x - \sin x}{x^3} \cos \frac{x}{2} dx$.

(c) Obtain the Fourier expansion of $f(x) = x \sin x$ as a cosine series in $(0, \pi)$. (06)

Hence show that $\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots \dots \dots \infty = \frac{\pi-2}{4}$.
